

STAT 207 Project 01 - Housing Prices

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Contribution Report

One person from your team should submit this assignment to Gradescope. After submitting, add your team members to the submission.

1. Amaya Hewamanage (amayash2) completed parts 1&3.
2. Maya Mazur (mimazur2) completed parts 2&4
3. We both worked on part 5 together. We both reviewed the full document before submission.

Group Work Reflection

Reflecting on your experience working as a group, how do you think the project went? What were the most effective things your group did to work together? What were the least effective things your group did to work together? What have you learned about group work from this project? What might you change about your group's process or dynamics for the next project? Did you face any substantial challenges in working together? If so, please explain any challenges that you had.

Our project went well, we both communicated with each other especially on topics of confusion. We also divided up the parts equally and fairly and communicated deadlines with each other. We did not face any challenges or difficulties working with each other.

1. Introduction

The reason why we choose this to explore is because we were curious about what factors affect the price of a house. We wanted to explore whether a specific characteristics specifically the city in which the house is located in, is associated with a house's sale price. Our research question is **how does the city of a house correlate to the sale price?** This question is important because location is a big factor in determining to buy a house or not. We wanted to see if understanding how location affects housing prices could help buyers compare and the differences in housing prices across locations. This is also useful for people selling their homes because they are able to understand how much the sale price of the home is affected by the location of their house. By exploring this question, we can understand how different cities affect the sale price of the homes in the city. This

is helpful information for someone who may be interested in buying a house in a different city and wants to understand which cities on average have more expensive houses vs. cheaper housing.

In [13]:

#Imports here
import pandas as pd
df= pd.read_csv('Housing.csv')
df

Out [13]:

	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront
0	2014-05-02	313000.00	3.0	1.50	1340	7912	1.5	0
1	2014-05-02	2384000.00	5.0	2.50	3650	9050	2.0	0
2	2014-05-02	342000.00	3.0	2.00	1930	11947	1.0	0
3	2014-05-02	420000.00	3.0	2.25	2000	8030	1.0	0
4	2014-05-02	550000.00	4.0	2.50	1940	10500	1.0	0
...	...	...	...	...	...	...	...	...
4595	2014-07-09	308166.67	3.0	1.75	1510	6360	1.0	0
4596	2014-07-09	534333.33	3.0	2.50	1460	7573	2.0	0
4597	2014-07-09	416904.17	3.0	2.50	3010	7014	2.0	0
4598	2014-07-10	203400.00	4.0	2.00	2090	6630	1.0	0
4599	2014-07-10	220600.00	3.0	2.50	1490	8102	2.0	0

4600 rows x 18 columns

2. Dataset Discussion

We found our data through the Kaggle database (<https://www.kaggle.com/datasets/debayank2024/house-price-prediction/data>) and we first accessed the data on September 27. While no formal codebook is included, there is a brief description on the data which includes variable and dataset descriptions. The data was originally collected from public real estate transaction records and property assessment data with a focus on properties in Washington.

The unit of observation for our data is each individual house. The column that helps identify a specific house in the dataset is the "street" column which gives the address of each house. Our two variables of interest are the city and the price of the home. The city is the categorical variable and is found in the dataset as "city". The price is the quantitative variable and is found in the dataset as "price". A limitation in our data is that this dataset focuses mainly on areas in Washington. If we wanted to expand our findings to the United States, we would not be able to since we are not provided with other states in the sample. In addition, there are some variables not included in the dataset that may be important to further understand the pricing of a home. For example, a variable for nearby public transport, a business hub, or a school may be interesting to analyze.

In [14]: `df.head(6)`

Out [14]:

	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view
0	2014-05-02	313000.0	3.0	1.50	1340	7912	1.5	0	0
1	2014-05-02	2384000.0	5.0	2.50	3650	9050	2.0	0	4
2	2014-05-02	342000.0	3.0	2.00	1930	11947	1.0	0	0
3	2014-05-02	420000.0	3.0	2.25	2000	8030	1.0	0	0
4	2014-05-02	550000.0	4.0	2.50	1940	10500	1.0	0	0
5	2014-05-02	490000.0	2.0	1.00	880	6380	1.0	0	0

In [15]: `#This dataset contains 4600 rows and 18 columns
df.shape`

Out [15]: (4600, 18)

3. Dataset Cleaning

Missing Variables:

Our data frame does not have any missing values that are explicitly identified by python as NaN, after checking for other possible missing values we found variables that contained 0s as the missing value. For the sale price (price) variable we found 49 missing values with a value of 0, this could mean that the house has not been on the market in a while and there is no recent data. For the city variable there are no missing values found. We are choosing to drop the 49 rows with houses that do not have a sale price because we are specifically interested in seeing how much the city has an effect on the sale price, so we need a valid value for the sale price for our analysis. Keeping these values will skew the data for the cities that have houses with a sale price of 0. This

decision allows us to compare the houses with sale prices and their cities without worrying about the effect of houses that don't have a sale price of 0 instead of an unknown sale price.

For the rest of the variables like, waterfront, view, sqft_basement, and the yr_renovated that had the missing variable 0, we suspect that it means that it is not applicable to the house and the house does not have this quality or characteristic. For example if the house has a value of 0 for the waterfront, we suspect that it means that the house does not have a waterfront. We found 49 price_per_sqft observations with missing values we believe that this is correlated to the 49 missing sale price variables, because the price per square foot cannot be calculated without the sale price.

Outside of our variables we found two unusual observations which are missing values for the bedrooms and bathrooms are a red flag. These values would be concerning if we were analyzing the bathroom or bedrooms, and we would choose to drop these two data points because a house must have at least 1 bathroom and 1 bedroom. However since we are specifically focusing on the city and its relationship to the sale price we will choose to ignore these two observations because it will not directly affect our analysis.

Uncommon Variables:

For our categorical variable, city, we found 42 unique categories by using value_counts(). Seattle had the most observations while Ingle-Finn Hill, Snoqualmie Pass, and Beaux Arts Village all had only 1 observation. We decided to keep these uncommon values rather than combining these observations because each observation represents that city. If we were to get rid of these data points we would eliminate the city from our analysis. If we were to combine the cities with less than 5 houses then it would remove the specific location data which we are analyzing, and would there for have no significance on how location relates to sale price. While we are keeping these values, we understand that having such few observations for some cities is a limitation of our analysis, and our analysis for these cities will be an incomplete picture and will not fully or accurately represent the houses in that city and their relationship with housing prices. So we will not make any observations about cities with such small observation numbers, because any analysis of these cities will not be able to reflect the market of the city properly.

For our quantitative variable, price, we used describe() to identify uncommon values. We found that the range of houses prices is 7800 to 26590000. While the maximum and mininum values are quite extremes when compared with the median value, we cannot write them off right away. We are not going to remove these values because having an unusually high or low value does not necessarily mean that there is an error with the value. These values may be accounting for other factors that are not described in the dataframe, like house condition or property size which are not the focus of our analysis. We will keep these observations rather than remove them. We will be aware of the influence these values will have when we perform our data analysis of the relationship between the city and sale price.

```
In [16]: df.isna().sum()
```

```
Out[16]: date           0
         price          0
         bedrooms       0
```

```

bathrooms          0
sqft_living         0
sqft_lot            0
floors              0
waterfront          0
view                0
condition           0
sqft_above          0
sqft_basement       0
yr_built            0
yr_renovated        0
street              0
city                0
statezip            0
price_per_sqft      0
dtype: int64

```

```
In [17]: df["city"].value_counts()
```

```

Out[17]: city
Seattle          1573
Renton            293
Bellevue          286
Redmond           235
Kirkland          187
Issaquah          187
Kent              185
Auburn            176
Sammamish         175
Federal Way       148
Shoreline         123
Woodinville       115
Maple Valley       96
Mercer Island      86
Burien             74
Snoqualmie        71
Kenmore            66
Des Moines         58
North Bend         50
Covington          43
Duvall             42
Lake Forest Park   36
Bothell            33
Newcastle          33
Tukwila            29
Vashon             29
SeaTac             29
Enumclaw           28
Carnation          22
Normandy Park      18
Fall City          11
Clyde Hill         11
Medina             11
Black Diamond       9
Ravensdale         7
Pacific            6
Algona             5
Yarrow Point        4
Skykomish           3
Preston            2

```

```

Milton                2
Inglewood-Finn Hill   1
Snoqualmie Pass       1
Beaux Arts Village    1
Name: count, dtype: int64

```

```

In [18]: dfmissing = pd.read_csv("Housing.csv",na_values =
    ["N/A","NA","Unknown","unknown","0"," "])
dfmissing.isna().sum()

```

```

Out[18]: date                0
price                49
bedrooms             2
bathrooms            2
sqft_living          0
sqft_lot             0
floors              0
waterfront          4567
view                4140
condition            0
sqft_above           0
sqft_basement        2745
yr_built             0
yr_renovated         2735
street              0
city                0
statezip            0
price_per_sqft       49
dtype: int64

```

```

In [19]: df=df[df.price !=0]

```

```

In [20]: df["price"].describe()

```

```

Out[20]: count    4.551000e+03
mean    5.579059e+05
std    5.639299e+05
min    7.800000e+03
25%    3.262643e+05
50%    4.650000e+05
75%    6.575000e+05
max    2.659000e+07
Name: price, dtype: float64

```

4. Summarizing Data

Our research question is how does the city of a house correlate to the sale price?

```

In [28]: import matplotlib.pyplot as plt
median_price=df.groupby('city')
['price'].median().sort_values(ascending=False)

```

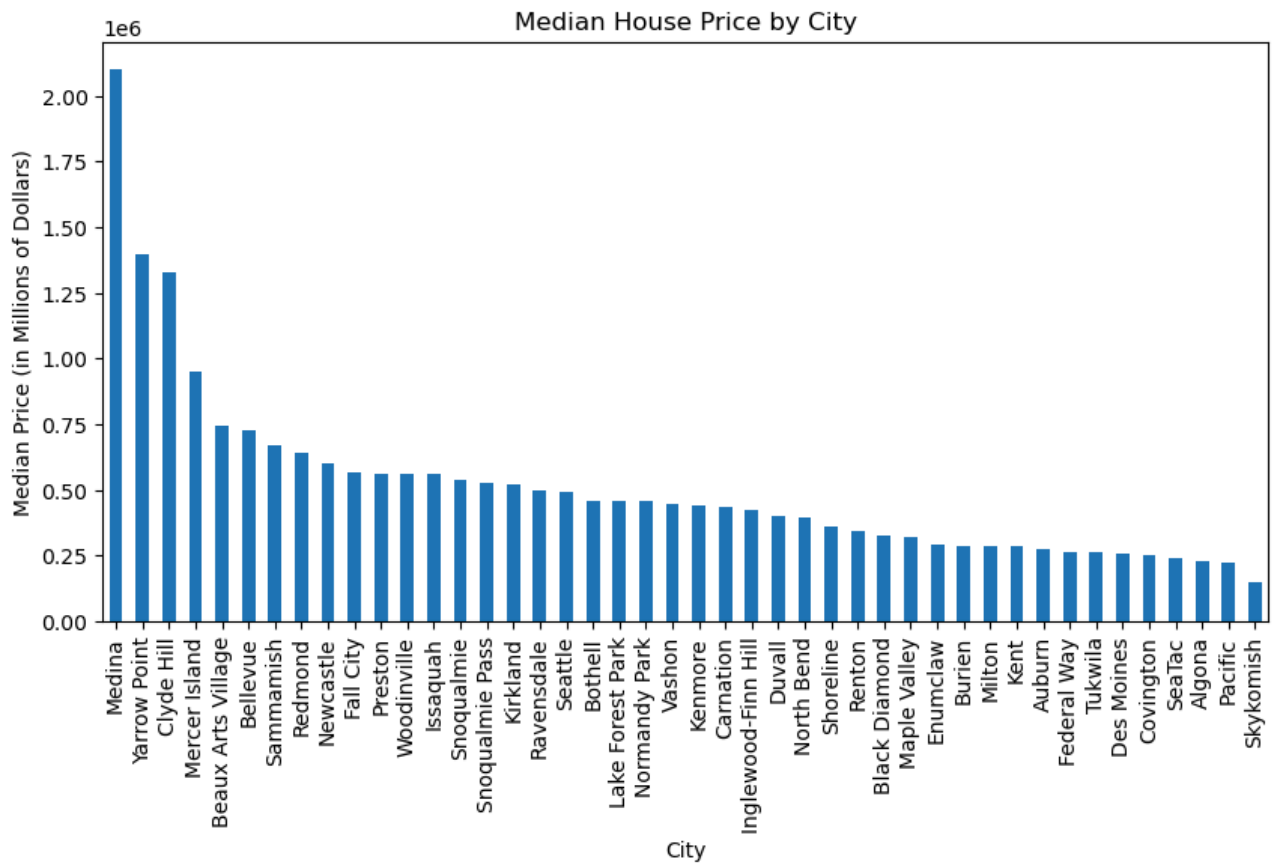
```

In [30]: median_price.plot(kind='bar',figsize=(10, 5),xlabel='City',ylabel=
'Median Price (in Millions of Dollars)',title='Median House Price by

```

```
City')
```

```
Out[30]: <Axes: title={'center': 'Median House Price by City'}, xlabel='City',
ylabel='Median Price (in Millions of Dollars)'>
```



```
In [29]: df['price'].describe()
```

```
Out[29]: count      4.551000e+03
mean      5.579059e+05
std       5.639299e+05
min       7.800000e+03
25%      3.262643e+05
50%      4.650000e+05
75%      6.575000e+05
max      2.659000e+07
Name: price, dtype: float64
```

The bar graph shows that the median house prices vary across the different cities in the dataset. Medina, Yarrow Point, and Clyde Hill, have a much higher median house price than the majority of the other cities. Most of the cities are in the lower-middle part of the chart, ranging from 0.25 million to 0.75 million. This shows that there is a noticeable difference in pricing depending on where the house is located. However, some cities don't have as many observations, which could lead to the median price not accurately representing the house market for those cities.

The numerical summary shows us that the distribution of individual house prices is right-skewed. The mean house price is around 0.56 million, while the median is 0.46 million. Since the mean is higher than the median, this tells us that the high houses prices are pulling the mean, resulting in right-skewed data.

Removing the 0\$ prices resulted in the minimum house price as 7,800 and the maximum as 26.59 million. These are unusually low and high values compared to the rest of the dataset, but as mentioned before, these high or low prices do not necessarily mean that the observation was an error.

These results also tell us that the median of 0.46 million is a better representation of the middle data as it is less affected by the extremely high prices. Overall, the results help answer our research question, "how does the city of a house correlate to the sale price?" The differences in median house prices show that the city is associated with sales prices in the dataset. Some had a much higher average sales price, suggesting that location is important to consider when comparing house prices.

5. Conclusion



Our research question was, how does location correlate to the sale price of the house. Before analysis, we cleaned our data and removed 49 observations with missing values for the price because we needed valid values for the price variable. We only found missing values for our price variable, there were no missing values for the city variable. There were missing values for other variables, but we decided to ignore them in this situation because we are not currently concerned with those variables and their values. We also compared the median sale price across different cities using the bar graph and summary statistic. Our results showed us that the house sale price differ across cities, suggesting that there is some sort of association with location and price.

A challenge we faced was how we should interpret the "price" column. We were aware of the extreme outliers but were not sure if it would be meaningful to help answer our question or not. We decided that the 0 price values would not help us understand how price and location are related so we filtered those values out. A limitation in our analysis was that for some cities, there were only a handful of observations in the data frame. Due to this, we were unable to accurately make a conclusion about the relationship between the city and the sale price. Another limitation is the fact that this dataframe only contains data from Washington; we are unable to apply any conclusions concerning the entirety of the US. Some important contextual information is that this data frame is a very very small sample size compared to the population of the whole of Washington state, so any claims made from this dataframe can not be taken completely as a statistic.

Our analyses sets up for further research on future questions. A question we may want to explore include: what are the factors that led to some cities having a higher average than others? We could bring in other variables from the dataframe and decipher the effect they have on the pricing. In the future, someone may want to expand this analysis to another state, or to one city and analyze the prices in different parts of the specific city.

AI Acknowledgement

Our course policy is that you should write all of your own interpretations and other narrative answers (phrases or sentences) yourself without the assistance of AI.

You may use AI to help guide your code, although you should write all of your own code yourself (not copy-paste from another source) and you should cite your use of AI. I would encourage you to try to generate any necessary code yourself first using course resources and using AI as a debugging tool if/when you reach an error that you can't figure out or to help you perform any coding tasks that are more advanced than we've demonstrated during class.

Did you use AI on this assignment? Did you use other resources outside of our course-provided resources on this assignment?

No

If you used AI or other resources, answer the following questions to cite your usage.

- Which AI and/or resources did you use?
- What prompts did you ask it?
- How did you integrate the responses into your assignment? Specifically, which questions or parts are associated with this usage?

Note: answering these three questions are enough for our course but may not be enough for a different course or context.